Math Inquirers Primary 1: Scheme of Work

Holiday program for ages 6 to 8 (builds on Primary 1 maths) • 10 lessons of 90 minutes • 2 lessons a week for 5 weeks • live online, up to 8 children

Approach. Enrichment, not extra drill: non-routine puzzles on topics chosen to complement the school syllabus. Lessons 1 to 9 follow the same routine, in minutes of the 90: Opening (2 to 4), warm-up (6), sneak-peek (4), explore (14 to 16), mini-lesson (8 to 11), brain break (4), worksheet (30 to 32), talk time (7), exit ticket (6), look-back (3), and flex time (2). Lesson 10 has its own plan. The time plan under each lesson is the one in the tutor guide.
Habits. Notice it. Test it. Explain why.
Coverage. This is enrichment. It does not teach time, multiplication and division, or 2-digit addition and subtraction within 100, so parents should not expect it to cover the whole Primary 1 syllabus.

Lesson	Title and school topic	Objective	Skills and heuristics practised	Exit ticket
1 Week 1A	**Count with the Frog** Number Patterns	Skip count in 2s, 3s and 5s; find and state an add or take-away rule; notice numbers that both counting patterns share.	Pattern spotting, rule finding, generalising	Next two numbers: 7, 10, 13, 16, __, __. What is the rule?
	Time plan (minutes): Opening **4** · Warm-up **6** · Sneak-peek **4** · Explore **14** · Mini-lesson **10** · Brain break **4** · Worksheet **30** · Talk time **7** · Exit ticket **6** · Look-back **3** · Flex time **2**			
2 Week 1B	**Number Bond Inquiry** Number Bonds	Know the bonds of 10, use them to find bonds of 20, balance equations, and reason about missing numbers.	Structure of numbers, systematic listing, reasoning about equality, guess and check	10 = 6 + __ 20 = 13 + __
	Time plan (minutes): Opening **2** · Warm-up **6** · Sneak-peek **4** · Explore **14** · Mini-lesson **10** · Brain break **4** · Worksheet **32** · Talk time **7** · Exit ticket **6** · Look-back **3** · Flex time **2**			
3 Week 2A	**Let's Plan the Fruit Party** Picture Graphs	Read and build a picture graph, compare and total data, and judge what a class vote can and cannot tell us.	Data reading, comparing, sampling awareness, weighing other factors	Dogs 6, cats 4, fish 3. How many pets altogether?
	Time plan (minutes): Opening **2** · Warm-up **6** · Sneak-peek **4** · Explore **14** · Mini-lesson **11** · Brain break **4** · Worksheet **31** · Talk time **7** · Exit ticket **6** · Look-back **3** · Flex time **2**			
4 Week 2B	**Pattern Inquiry** Repeating and Growing Patterns	Extend a repeating pattern and predict far places using the pattern core; extend a growing stick pattern and work backwards from a total.	Core identification, prediction by multiples, tables, working backwards	Circle, square, circle, square ... What is the 9th shape?
	Time plan (minutes): Opening **2** · Warm-up **6** · Sneak-peek **4** · Explore **14** · Mini-lesson **11** · Brain break **4** · Worksheet **31** · Talk time **7** · Exit ticket **6** · Look-back **3** · Flex time **2**			
5 Week 3A	**Mystery Number** Working Backwards	Solve "I think of a number" problems by working backwards, find a number machine rule, and solve more-than and fewer-than stories.	Inverse operations, rule finding, bar-model thinking, logical reasoning	I think of a number. I add 6. I get 15. What is my number?
	Time plan (minutes): Opening **2** · Warm-up **6** · Sneak-peek **4** · Explore **14** · Mini-lesson **10** · Brain break **4** · Worksheet **32** · Talk time **7** · Exit ticket **6** · Look-back **3** · Flex time **2**			
6 Week 3B	**Coin Combos** Money	Find all the ways to make an amount, work systematically and use reasoning to rule out cases. (Paying in dollars is an extra on the Think harder page.)	Systematic listing, comparing, working backwards, justifying "there are no more"	Show one way to make 30¢ with 20¢, 10¢ and 5¢ coins.
	Time plan (minutes): Opening **2** · Warm-up **6** · Sneak-peek **4** · Explore **14** · Mini-lesson **10** · Brain break **4** · Worksheet **32** · Talk time **7** · Exit ticket **6** · Look-back **3** · Flex time **2**			
7 Week 4A	**Shape Safari** 2D Shapes	Name and describe 2D shapes by sides and corners, count shapes in a picture, count triangles in a figure, and compose shapes.	Classifying, describing, systematic counting, spatial reasoning	A shape has 6 sides. How many corners does it have? What is the shape called?

Lesson	Title and school topic	Objective	Skills and heuristics practised	Exit ticket
	Time plan (minutes): Opening **2** · Warm-up **6** · Sneak-peek **4** · Explore **16** · Mini-lesson **8** · Brain break **4** · Worksheet **32** · Talk time **7** · Exit ticket **6** · Look-back **3** · Flex time **2**			
8 Week 4B	**Logic Lab** Logic and Sorting	Order things from clues, sort using two rules with a Venn diagram, solve a logic grid, and explain an odd one out.	Deduction, classification, elimination, justifying	Ben is shorter than Mei. Mei is shorter than Ali. Who is the tallest?
	Time plan (minutes): Opening **2** · Warm-up **6** · Sneak-peek **4** · Explore **14** · Mini-lesson **10** · Brain break **4** · Worksheet **32** · Talk time **7** · Exit ticket **6** · Look-back **3** · Flex time **2**			
9 Week 5A	**Measure It!** Length	Compare and order lengths, measure with equal units and with a centimetre ruler (also when the object does not start at 0), and change a length from one unit to another in both directions.	Comparing, measuring, unit reasoning, estimating	A ribbon lies from the 3 cm mark to the 9 cm mark on a ruler. How long is it?
	Time plan (minutes): Opening **2** · Warm-up **6** · Sneak-peek **4** · Explore **15** · Mini-lesson **10** · Brain break **4** · Worksheet **31** · Talk time **7** · Exit ticket **6** · Look-back **3** · Flex time **2**			
10 Week 5B	**Puzzle Day** Mixed Puzzles	Use patterns, bonds, graphs, money, shapes, logic, working backwards and measuring in new puzzles, and explain the thinking.	Applying and connecting ideas, perseverance, explaining	My favourite lesson was ____ because ____.
	Time plan (minutes): Opening **2** · Warm-up **4** · Sneak-peek **4** · Explore **18** · Mini-lesson **9** · Brain break **4** · Worksheet **30** · Talk time **8** · Exit ticket **8** · Look-back **3**			

Assessment and progress

Each lesson has a 2-page Start here worksheet, then Think harder, Go further and an Investigation with a Look-back (printed for every child, who may choose any of them at any time; nobody is placed in a group), an exit ticket and an Inquiry Journal stamp. Tutors record S (secure), D (developing) or H (needs help) for each child on the progress tracker in the tutor guide. Parents receive a short note each week and the Weekly Dose of Thinking Math practice pack.

Materials

Printed student workbook with cut-outs, pencil, scissors and paper; a laptop or tablet with a camera. Tutors use the slides for each lesson and the tutor guide scripts.